Practical 3B: Mineralization sequence in an ore sample

Economic 'ore-forming' minerals are those from which we extract many of our necessary metals. Often these minerals occur as hydrothermal products associated with late-stage igneous activity. Because these magmas are emplaced at near-critical conditions of water, exsolved fluids are able to fracture the rock by fluid expansion, and mineralized fractures form. These are fossilized fluid pathways and are referred to as veins. Fluid composition determines the ore mineralogy, but it may change over time, leading to multiple generations of veins.

This exercise is designed to familiarize you with the geological history of an ore deposit. By determining cross cutting relationships and mineralogy in a sample, you can reconstruct the evolution of mineralizing fluids.

1. Examine the rock slab or image. You will notice that several veins, of different color, cut the matrix (host or country rock).
2. Based on your visual inspection, determine the number of different veins in the sample, based on color.
3. Determine the age of different veins using cross cutting relationships. List the veins by color from oldest to youngest.
4. Based on mineral color, determine the mineralogy.